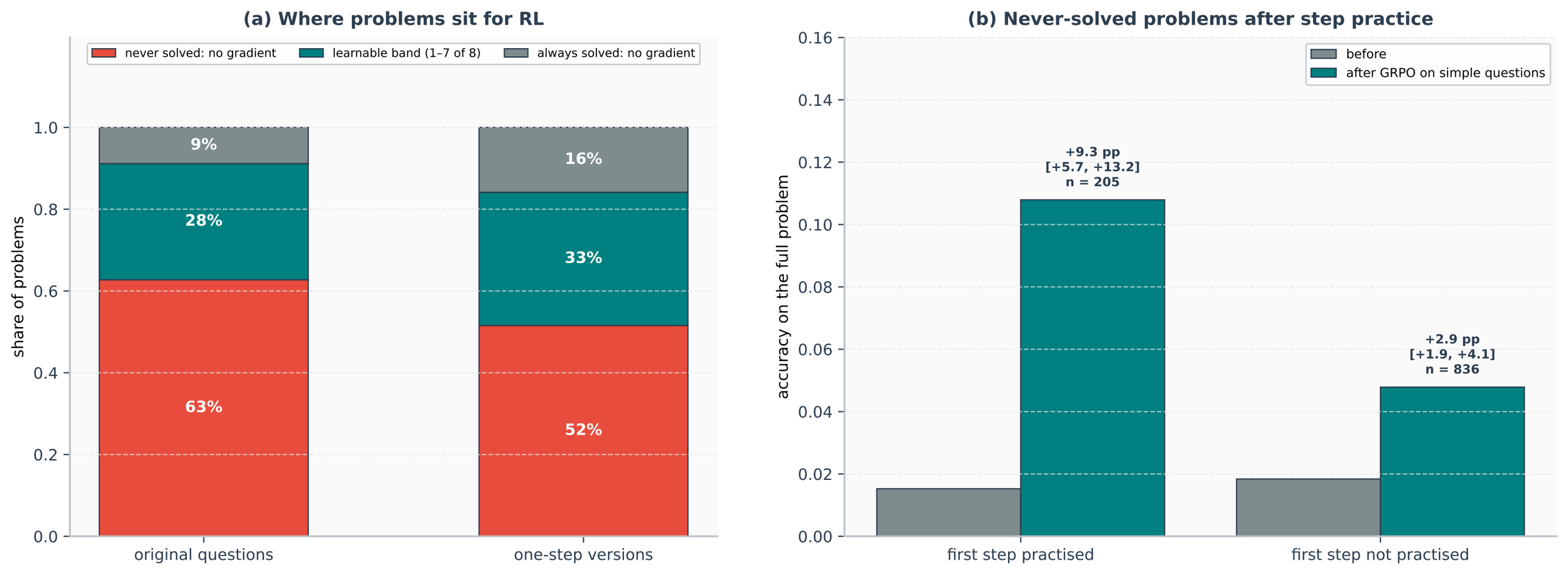
(a) Where problems sit for RL
never solved: no gradient
learnable band (1–7 of 8)
always solved: no gradient
share of problems
1.0
0.8
0.6
0.4
0.2
0.0
9%
28%
63%
16%
33%
52%
original questions
one-step versions
(b) Never-solved problems after step practice
before
after GRPO on simple questions
accuracy on the full problem
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
+9.3 pp
[+5.7, +13.2]
n = 205
+2.9 pp
[+1.9, +4.1]
n = 836
first step practised
first step not practised